

Title: Asymmetric reverting and volatility in cryptocurrency markets

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Abstract

The study employed the asymmetric nonlinear smooth transition generalized autoregressive conditional heteroskedasticity (ANST-GARCH) model to investigate the asymmetric responses of mean reversion and volatility in cryptocurrency markets. The data set is a daily observation period spanning from 20th April 2013 to 21st February 2023. The study found strong evidence for the ANST-GARCH model as well as considerable asymmetries concerning the behaviour of investors who overreact to certain cryptocurrency market news. The study also revealed that asymmetric volatility is crucially significant in cryptocurrency markets and that investors can implement risk management techniques and trading strategies to mitigate the effect.

Keywords: Cryptocurrencies; unit root; asymmetric; volatility; ANST-GARCH model

JEL Classification: C4; G14; G15

1.0 Introduction

Uncertain movements in stock prices are hard to follow as fear and uncertainty impact market participants in the global financial markets. While today's macroeconomic models may not be able to precisely forecast cryptocurrency's economic and financial impact. The concepts of cryptocurrencies are based on a blockchain technology but cryptocurrency trading is characterized by speculation (Bindseil and Schaaf, 2022). Cryptocurrencies trading are extremely speculative and volatile (Katsiampa, 2017) and prices are more volatile compared to prices of commodities like gold (Yermack, 2013). Cryptocurrencies are based on a blockchain technology, most important innovations in the area of FinTech (Siu and. Elliott, 2020). Volatility is a crucial variable for assessing the circumstances of financial and capital markets towards decision-making by investors and financial regulators. Bouri et al. (2019) investigated herding behaviour in cryptocurrencies and found that herd investing leads to inadequate portfolio diversification and risk and that investors holding only cryptocurrencies face further risk. Cryptocurrencies have exploded over the last few years, but many still doubt precisely what they are, how they work, why are there so many cryptocurrencies, and how to trade them successfully. Notwithstanding, cryptocurrencies have become so valuable in many economies in the world. Cryptocurrency is having a colossal impact on the conception of money as another kind of monetary currency for payment systems and money transfer payment in most economies in the world. These countries see the potential benefits that cryptocurrencies would bring in the growing liquidity needs for governments and improvements in pricing levels.

Cryptocurrency continues to become an innovation within the financial system, gradually becoming conventional as a digital asset investment, technological infrastructure and a sharp development tool, but this market is characterized by asymmetry volatility. Investors are skeptical as the variability of return is significantly volatile and problematic to forecast. Dourado and Brito (2014, p 4) indicated that "cryptocurrencies do not have central banks to regulate the money supply but rather their controlled are embedded in the peer-to-peer software". This means that cryptocurrencies do not have a physical central bank or regulating authority, but rather have a decentralized system to record dealings and production of a new deal. The asymmetric

volatility exemplified that negative return shocks developing from bad news wield significant effect on intensifying the future volatility compared to its equivalent positive return shocks developing from good news (Black, 1976; Campbell and Hentschel, 1992; Blasco et al., 2002). Nelson (1991) showed that asymmetric mean reversion exhibits negative return shocks, quickening a faster reversion compared to its equivalent positive return shocks. The effect is mostly that asymmetric deteriorations in stock prices are conveyed by larger upsurges in volatility than the decrease that conveys rising stock markets. Financial economics advocate that the stock market falls when volatility is high and the stock market rises when volatility is low, making stock returns negatively correlated with future volatility.

Prior studies have used stock market overreaction hypothesis to explain the asymmetric volatility. The study thus, used the reverting behaviour of cryptocurrency price is ascribed to investors revision of their expected returns to investigate the asymmetric volatility using the asymmetric nonlinear smooth transition generalized autoregressive conditional heteroskedasticity (ANST-GARCH) model.

Given these caveats, it is argued that the presence of asymmetric reverting and volatility in cryptocurrencies can be explained from several perspectives. This paper examines the price implications of asymmetric reverting and volatility by investigating whether cryptocurrency returns show the presence of asymmetric reverting and volatility. The objective is to test for the presence of asymmetric reverting and volatility when asymmetric reverting magnitudes and volatilities are extremely expected to develop. The motivation of this paper can be extended by the theory of negative leverage by Black (1976). Black's (1976) leverage effect which posit that a negative stock return novelty upsurges the financial leverage of a firm since the value of equity; reduces when debt is incorporated, which, in turn, leads to higher equity return volatility. Black (1976) was the first to propose the asymmetric volatility and ascribed this novelty to the leverage effect.

The ANST-GARCH technique was applied across the top 17 cryptocurrencies involving about 80 per cent of the whole cryptocurrency market capitalization. A significant number of empirical studies have earlier used the GARCH family to test for asymmetric volatility across different financial markets and how it affected different results. Nam et al. (2002) used the ANST-GARCH model to justify asymmetric mean-reversion and contrarian profits. Nam et al. (2003) also employed the ANST-GARCH model to explain the asymmetrical return reversion in Pacific-basin markets in the short horizon. Yang and You (2003) investigated the asymmetric volatility of Asian Pacific markets using the exponential general autoregressive conditional heteroscedasticity (EGARCH) approach. In a recent empirical study, Stavroyiannis (2017), and Colucci (2018) used the generalized autoregressive conditional heteroscedastic (GARCH) model and related models to investigate the characteristics of Bitcoin returns, and they claimed that bitcoin returns are characterized by the volatility perceived in the Bitcoin market.

Policymakers acknowledge that there is a compelling risk that cryptocurrency-associated financial risks are not fully reflected in crypto assets returns. Behavioural finance in the cryptocurrency market, where participants make systematic mistakes when they invest, create persistent inefficiencies in the market. Behavioural finance as a challenge to the cryptocurrency

market and how psychological factors can affect investors' decisions. Often overconfident after a few easy wins leads to an underestimation of risk on the part of investors. Cryptocurrency markets are highly dependent on social networks such as Twitter, Skip, Reddit, Quora and other social platforms. These individual investors want to get rich quickly, boast of their wealth and make an abnormal return. The magnitude of abnormal returns can depend on the volatility of the returns and evidence in support of the volatility feedback effect (see, e.g., French et al., 1987).

The concepts and practical applications of cryptocurrency trading, technology, the blockchain origins and its relationship with central banks and policymakers remain a mystery. Policymakers seem not to fully understand these complex technologies to analyze coherent challenges and opportunities linked with them in the business environment. The most prevalent one is Bitcoin, developed in the innovative seminal paper by Nakamoto (2009). Crypto asset remains strange in the contentious cryptocurrency market that presently irritates many individual investors' interest. Cryptocurrencies exhibit unexpected returns and tremendous volatility with no apparent substantiating news. Bouri et al. (2017a) studied the bitcoin return-volatility relationship around the market crash of 2013 and indicated that serial correlation in Bitcoin return stimulates enormous consideration from portfolio investors owing to little correlation. Huynh et al. (2020) investigated the transmission process measured by informational flows from the US and European crude oil markets to many cryptocurrencies using the entropy transfer methodology and Granger's (1988) causality method. The study found that most cryptocurrencies are interconnected with the crude oil market.

The study made several contributions to literature by extending the asymmetric reverting and volatility persistent in 17 cryptocurrencies for the period covering 20th June 2013- 21st February 2023, using the ANST-GARCH model. The study differs from previous studies for several reasons. First, it sought to determine the price implications of asymmetric reverting and volatility by investigating whether cryptocurrency returns show the presence of asymmetric reverting and volatility. Secondly, volatility is usually greater after the stock market tumbles compare to when it increases, thus cryptocurrencies returns are negatively correlated with potential volatility. Bindseil and Schaaf (2022) justified that Bitcoin only creates negative value for investors. Thirdly, we examined the pricing of cryptocurrency with the aim of capturing the speed of transition between volatility regimes. The bigger the value of the parameter the quicker the transition between volatility regimes. Lastly, there is no known study examining asymmetric reverting and volatility in cryptocurrencies using an ANST-GARCH model. An ANST-GARCH model is the chosen model measurement relation to the typical GARCH model which was employed by Stavroyiannis (2017), and Colucci (2018) to test the characteristic of Bitcoin returns. The model's innovation uses an ANST-GARCH model to explore the price implications of asymmetric reverting and volatility. This model has numerous innovation implications that link asymmetric reverting and volatility. The paper extended the application of this quantitative method to examine the asymmetric reverting and volatility of 17 cryptocurrencies using daily cryptocurrency data and utilizing diverse techniques from prevailing literature.

Our preferred model is the ANST-GARCH model as against GARCH model, because GARCH models is their unsuitability, where an asymmetric effect in normally detected and different volatility resulting to good and bad news. GARCH models presupposes deterministic volatility

based on past returns and conditional variances. Actually, it may not be correct. One key advantage of the ANST-GARCH model delivers an efficient and parsimonious means permitting a nonlinear requirement to capture an asymmetry in both conditional mean and variance procedures, resulting in positive and negative shock (Nam et al., 2002 and 2003). The objective is to indicate that using ANST-GARCH estimation is better than normal GARCH estimation. ANST-GARCH has improved on the estimation method.

The remainder of this paper is organized as follows. Section 2 presents the literature review and highlights both the contribution and applications of GARCH models. Section 3 describes the ANST-GARCH model and the data set. Section 4 presents the empirical findings. Finally, section 5 concludes and policy implication of the paper.

2.0 Literature review and hypotheses developments

The concepts of asymmetric volatility could be explained by reverting behaviour of stock prices. Wei (2018) indicated that returns from various cryptocurrencies exhibited signs of autocorrelations. The author showed that the information efficiency of cryptocurrency returns is stronger once the liquidity of cryptocurrency market is greater. However, this result is different from others. Trimborn et al. (2019) showed that cryptocurrencies have a lower of liquidity compared to traditional assets. In 2022 fear and uncertainty impacted market participants looking to investors to navigate volatility and new possibilities in the global market. After months of price hikes global price volatility for fuel increased sharply, led by an increase in diesel and petrol in the global market. The intersection of complex factors, from climate change to Covid 19 pandemic to conflict (Russia-Ukraine war) led to climbing costs and severe supply chain strain, resulting in have heavily impacted the world economies and these exogenous shocks hurt the financial markets. The biggest cryptocurrency market Bitcoin exhibited tremendous speculative nature which makes the cryptocurrency market extremely volatile and may lead to asymmetric volatility (Baur et al., 2018), however, Bitcoin was at a new all-time high in late 2021. This contributed to millions of short liquidations over the past weeks as prices pushed higher.

Information accessibility in the stock market and uncertainty between crypto trader's influence how investors read between the lines and respond to the financial information published by firms. The formation uncertainty on cryptocurrencies can have a significant impact on diverse economies and financial assets. Aslan et al. (2011) are of the view that bigger companies tend to have smaller information asymmetry since more investors are negotiating the company shares. The authors assert that assets with a greater volume of private information tend to be riskier, and that leads to a higher expected return. Consistent with common intuition, large negative stock returns are appearing to be more than large positive stock returns and therefore indicate negatively skewed. Recent empirical and theoretical studies have examined the bitcoin options derivatives futures markets reaching different conclusions (Pagnotoni, 2019; Shi and Shi, 2019; Alexander et al., 2020). Siu and Elliott (2020) showed that the effect of conditional heteroscedasticity on Bitcoin prices is quite important. The overall risk of the portfolio can be minimized if investors add an asset of bitcoin to their portfolio (Guesmi et al. 2018).

More importantly, given information asymmetry in emerging markets investors are prone to volatility and to respond asymmetrically to extreme market movements. A deep understanding of how cryptocurrency works in the global financial market and how investors make up their returns is fascinating. Boyer et al. (2006) investigated crisis spread and found that greater co-movement during high volatility periods in emerging stock markets is predominantly due to contagion but not changes in economic fundamentals in the market. The rate of quick expansion of blockchain and cryptocurrency in the financial market coupled with investors' interest in these highly volatile assets in making their investment decisions are appropriately priced. There is empirical research on the price discovery and valuation in cryptocurrency markets. Lehner et al. (2018) used the GARCH model for studying cryptocurrency valuation models. Jin et al. (2019) used a multivariate GARCH (MV-GARCH) model to investigate the relationship between Bitcoin and oil and found that Bitcoin prices are prone to price fluctuations. Liu (2018) indicated that investors obtain substantial returns in cryptocurrencies by diversifying across different markets. Kaiser (2018) found no support for robust and calendar effects on cryptocurrency returns when he investigated the seasonality effect on cryptocurrencies. His result suggested that volatility and trading volume are normally lower during the weekends and the month of January. Cubillos-Rocha et al. (2019) examined exchange rate contagion and found evidence for asymmetric dependence throughout the periods of currency appreciations. Kwapien et al. (2021) indicated that the cryptocurrency market has slowly developed more solidly from a topological viewpoint.

Kallinterakis (2019) showed that in cryptocurrency markets there is tremendous noise and volatility linked to the financial markets. Mensah and Premaratne (2017) also found evidence for asymmetric dependence in the Asian banking sector. Understanding how cryptocurrency works in the financial system and the economy is core to fulfilling investment goals in the financial market. Despite investors' interest in cryptocurrency remaining high, the cryptocurrency market could plunge due to extreme market conditions. Financial analysts think that the volatility in the market is high so they advocate retaining cryptocurrency investment to less than 5% of one's portfolio to keep started. Cryptocurrencies, like any other stock commodity, are likely to suffer from macroeconomic circumstances like increasing interest rates, inflation, and the Russia-Ukraine war. Bouoiyour and Selmi (2017) indicated that Bitcoin is a high-risk investment. They also illustrated that the Bitcoin market is not developed adequately to be incorporated into global financial markets. It is unclear how these cryptocurrencies will develop in the long run and what their decisive impact on the financial markets will be. One of the significant issues Some market analysts think that cryptocurrency would be directed and controlled. As indicated by Marcus Sotiriou a market analyst at digital asset broker Global Block "after the catastrophic events that have unfolded in the crypto market over the past few weeks, it is clear that stringent regulation could arrive soon,"

Given the empirical and theoretical studies on risk and asymmetric volatility (Baur et al., 2018; Kallinterakis, 2019; Cubillos-Rocha et al., 2019) the central banks, supervisors, policymakers and financial institutions have a major role to participate in achieving the objectives of the cryptocurrencies in financial markets. Consequently, how central banks are able to manage risk and asymmetric volatility may represent a realistically significance and essential issue. Several empirical research on asymmetric volatility abounded. The ANST-GARCH and related models

were applied to model an asymmetric volatility in financial market. While the empirical work of Yang and You (2003) examined asymmetric volatility through the EGARCH, Liau and Yang (2008) studied asymmetric volatility in the Asian financial crisis through the ANST-GARCH model. Nam et al. (2003) also used the ANST-GARCH model to explain the asymmetrical return reversion in Pacific-basin markets.

However, none of the previous studies examined the price implication of asymmetric volatility in cryptocurrency using the ANST-GARCH approach. Thus, this study's particular interest in examining the price implications of asymmetric volatility by exploring whether cryptocurrency returns show the presence of asymmetric volatility. The study also contributes to the literature by extending the asymmetric volatility persistent in 17 cryptocurrencies involving about 80 per cent of the whole cryptocurrency market capitalization. This is the first cryptocurrency study that has used the ANST-GARCH estimation method.

3.0 Methodology and data sets

3.1 Methodology

To estimate the model, the study employed the ANST-GARCH model due to Nam et al. 2003. The ANST GARCH model makes use of theories about asymmetry in both the conditional mean and the variance concurrently evaluated. This approach is to maintain parsimony within the model that captures the asymmetry in both return setbacks and return instability. Specifically, we follow Nam et al. 2003 ANST GARCH model which is stated as follows:

$$R_t = \mu + [\phi_1 + \rho_1 D(\xi_{t-1})]R_{t-1} + \xi_t, \quad (1)$$

$$h_t = [ao + a_1 \xi_{t-1}^2 + a_2 h_{t-1}] + [bo + b_1 \xi_{t-1}^2 + b_2 h_{t-1}] F(\xi_{t-1}), \quad (2)$$

where $F(\xi_{t-1}) = \{1 + \exp[-\gamma(\xi_{t-1})]\}^{-1}$ and $\exp$ denotes exponential function. R_t is the price of stock market return at t . R_{t-1} is the price of stock market return at $t-1$. ξ_t represents a new market shock at a time t and $\xi_t \sim N(0, \sqrt{h_t})$. $D(\xi_{t-1})$ denotes an endogenous dummy variable that takes on a value of one if $\xi_{t-1} < 0$ and zero otherwise. Consequently, serial correlation in returns is determined by $\phi_1 + \rho_1 D(\xi_{t-1})$ and thus makes sure that $[\phi_1 + \rho_1 D(\xi_{t-1})]$ is stationary. The functions of equations 1 and 2 can incorporate asymmetry in both conditional mean and variance procedures into the measurement and the influence of a return shock on both the conditional mean and the variance procedures is concurrently calculated.

In Equation 1, an asymmetric reverting condition exist $\rho_1 < 0$ which confirms asymmetric return reversals, a negative return shock associated with bad news has the greatest tendency of reverting magnitude, compared to a positive return shock triggered by good news. This means that negative returns show massive reverting displays relative to positive returns. In Equation 2, conditional variance processes show the volatility transition method for distinct volatility conditions. The model captures volatility by

$(a_1 + a_2) + (b_1 + b_2) * (1 + \exp[-\gamma(\xi_{t-1})])^{-1}$. Also $F(\xi_{t-1}) = (1 + \exp[-\gamma(\xi_{t-1})])^{-1}$ the logistic function. Once $F(\xi_{t-1}) = 1$, indicates a large positive shock and volatility is calculated as $(a_1 + a_2) + (b_1 + b_2)$. Thus, volatility is measured $(a_1 + a_2)$ once a large negative shock generates $F(\xi_{t-1}) = 0$. The asymmetric volatility is portrayed as $(b_1 + b_2) < 0$. The parameter $\gamma F(\xi_{t-1})$ influences the transition speed between diverse instability regimes and the transition speed between diverse instability regimes' upsurges γ .

3.2 Data set

To conduct the analysis, we use daily trading activities of 17 cryptocurrencies from coinmarketcap.com. The data set is a daily observation period spanning from 20th April 2013 to 21st February 2023. The daily frequency data for the 17 cryptocurrencies – bitcoin (BTC), Ethereum (ETH), litecoin (LTC), nem (XEM), dash (DASH), dogecoin (DOGE), monero (XMR), vertcoin (VTC), verge ((XVG), digibyte (DGB), Stellar (XLM), tether (USDT), maidsafecoin (MAID), ripple (XRP), bitcoin cash (DCH), cardano (ADA), and siacoin (SC). A extended period to conceal as much information was used and involved about 80% market capitalization of the cryptocurrency market. Before estimating the variables, all the return series were converted into natural logarithms. The Appendix 1 below provides each of the cryptocurrency name, market capitalization, circulating supply and date of issue.

4.0 Empirical results

4.1 Descriptive statistics

Table 1 presents the descriptive statistics of 17 cryptocurrencies in terms of daily trading activities. The mean log returns are positive except tether (USDT) which is negative. All 17 cryptocurrency variables show high volatility with Bitcoin (BTC) being the highest volatility. Bouoiyour and Selmi (2017) designate that Bitcoin is high-risk and volatile for investors. Skewness is mostly positive and statistically significant for cryptocurrencies. Only siacoin (SC) and tether (USDT) have negative and statistically significant skewness for the cryptocurrencies.

[Table 1, about here]

Excess kurtosis is significant for all cryptocurrencies and all variables have a heavy tail distribution. Overall, the presence of kurtosis stipulates evidence for volatility clustering and fat-tailedness (Joseph et al., 2020). As expected, the Jarque-Bera statistics demonstrate that the returns for all cryptocurrencies are not normally distributed. The Jarque-Bera for all the variables is statistically significant. The normality assumption is rejected, suggesting there is interconnectedness in the abnormal distribution of variables (Huynh et al. 2020)

4.2 Pearson correlation matrix

Table 2 depicts the Pearson correlation matrix between independent variables. Table 2 shows the correlation coefficient log returns of 17 cryptocurrencies. The correlation values between cryptocurrencies are low and with the highest estimated correlation being 0.62 between XVG and XMR, and the second highest correlation being 0.60 between BTC and ETH, respectively. Both correlations are positive and significant. The results show that cryptocurrencies are

negatively and positively correlated. From Table 2, none of the variables is extremely correlated, that is to say there are no multicollinearity.

[Table 2, about here]

4.3 Collinearity statistics

Table 3 shows the tolerance and variance inflation factor (VIF). As can be seen from Table 3, there is no multicollinearity for tolerance and the VIF. We used tolerance and VIF to verify multicollinearity. When the calculated tolerance is close to 1, it means there is tiny multicollinearity. A calculated tolerance close to 0 implies that multicollinearity may exist. The tolerance signifies the percentage of variance in certain predictors that cannot be described by the additional independent variables. The reciprocal of tolerance is recognized as the VIF (Oke et al., 2019; Senaviratna and Cooray, 2019). Jamal (2017) shows that when the calculated VIF values are greater than 5 or 10, it denotes that the associated regression coefficients are calculated due to multicollinearity.

[Table 3, about here]

4.4 Unit root tests

Before analyzing the results, the stationary properties of the data are presented in Table 4. The Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) tests were used for each set of data. This was to make sure the stationary time series data sets' results are unbiased and not spurious. The null hypothesis for each of the ADF and PP tests is the variable has a unit root. The ADF and PP statistics could not reject the null of a unit root in the (log) level of the univariate returns. The appropriate lag structures for the Schwartz Information Criterion (SIC) for ADF and Newey West were selected using the Bartlett kernel for the PP test. We perform the test on the level and the first difference of each series.

[Table 4, about here]

Performing the test on the first difference, all variables were stationary at the 1% significance level from the series. Again, performing the test on the level, all variables were stationary at the 1% significance levels, except for DOGE and SC for the ADF test and DOGE for the PP test which were at the 5% significance level from the series. Thus, cryptocurrency returns have a unit root, and this indicates that cryptocurrencies are integrated at $I(0)$.

4.5 ANST-GARCH results

In this section, the ANST-GARCH model is estimated. The empirical results are shown in Table 5. Maximum likelihood estimation to estimate the ANST-GARCH model was used to evaluate the most appropriate condition for the data. Diagnostic tests were also used for model appropriateness using the Ljung-Box test for the standardized squared residuals. We used sign and size bias tests due to Engle and Ng (1993) to determine whether an asymmetric model can be

considered satisfactory. It is important to note that the magnitude or size of the shock which affects the response of volatility to shocks is asymmetric or otherwise.

In both cases $\phi_1 > 0$ and $\rho_1 < 0$ are highly statistically significant at the 1% level across the entire cryptocurrency market. Thus, the results indicated that both $\phi_1 > 0$ and $\rho_1 < 0$, the serial correlation of the coefficient's estimated magnitudes were smaller for a past negative return shock than that of a past positive return shock (Liau and Yang, 2008). These results are consistent with Corbet and Katsiampa (2018) who found that negative returns decrease marginally and are smaller than the number of positive returns. The results showed an asymmetric reverting behaviour between negative and positive returns, with negative returns reverting more rapidly than positive returns. Consequently, in the rising market in cryptocurrency, one would expect more positive returns than negative returns.

The higher the ϕ_1 , more persistent the positive returns. The effect is stronger in magnitude in the BTC cryptocurrency. This result supports the theory that positive sentiment in the cryptocurrency markets has a positive influence on the price of cryptocurrency (Gurdgiev and O'Loughlin (2020). According to the authors (Gurdgiev and O'Loughlin, 2020), cryptocurrencies experience an upsurge in price when opinions toward their underlying technology, development and price dynamics are positive. Thus, the results of the coefficients $\phi_1 > 0$ are always positive and are significant and consistent with the findings of Liu (2018) discovered that investors can achieve considerable returns by diversifying across different cryptocurrencies.

This indicated that positive returns are continual and negative returns display a reverting fashion. This means that negative returns are more probable to revert to positive returns, even though positive returns are less probable to revert to negative returns (see e.g. Nam et al., 2003; Liau and Yang, 2008; Mozouz et al., 2009). The findings are consistent with the hypothesis that crypto asset investors are more sensitive to past bad news and are less likely to overreact to past good news (Mozouz et al., 2009). However, cryptocurrencies showing relatively high or low returns would be likely to conclude on the way of their returns.

The ANST-GARCH is likely to capture both conditional volatility and asymmetry that might influence the time series regression. The $(b_1 + b_2)$ coefficients which capture asymmetric volatility are mostly negative and significant across all cryptocurrency markets. The finding suggested that asymmetry in conditional volatility of the cryptocurrency market is weak. The asymmetric volatility is captured by $(b_1 + b_2) < 0$ and the asymmetric response of volatility to return shocks clutches in cryptocurrencies (Liau and Yang, 2008).

As expected, negative shocks generate the largest impact and influence future volatility in the cryptocurrency markets more than exhibit positive return shocks of similar magnitude. Several economic theories seek to explain the volatility in cryptocurrencies. A common finding across these studies Gronwald, 2014; Bouri et al., 2017a; Colucci, 2018; Katsiampa, 2017; Siu, 2019a indicated that a major problem for the empirical features of Bitcoin prices is the volatility detected in the Bitcoin market. In a related context, Osterrieder and Lorenz (2017) compared the volatility of global ten currencies with the volatility of Bitcoins and found that the volatility of

Bitcoins could be six to seven times higher. To reduce volatility in cryptocurrencies, investors could diversify against volatility in a commodity market

[Table 5, about here]

Nam et al. (2003) indicated that an asymmetric reverting condition exist $\rho_1 < 0$ which confirms asymmetric return reversals, that a negative return shock associated with bad news has the greatest tendency to revert magnitude, compare to a positive return shock triggered by good news. This means that negative returns show massive reverting displays relative to positive returns.

The presence of significant Ljung-Box test statistics for some cryptocurrencies (for instance DOGE and USDT) indicates potential issues with autocorrelation in the residuals. This could imply that the model may not be fully capturing the dynamics of the data. Significant values in the sign bias and size tests might indicate the model's varied ability to capture asymmetric responses to positive and negative shocks in the selected sample.

The coefficients for γ capturing the conditional variance are positive and significant for all cryptocurrencies. The conditional variance on cryptocurrency returns shows conditional asymmetry-leverage effects which are highly positive and significant for all cryptocurrencies. The leverage effect, this implied that the cryptocurrency market is volatile. The γ parameter measures the transition speed between volatility regimes regimes, a greater γ specifies an acceleration of the transition between volatility regimes regimes. The parameter γ that captures the transition speed between different volatility regimes and differs for the cryptocurrencies and higher for all return series. Kallinterakis (2019) found a substantial herding effect in cryptocurrencies.

4.6 Robustness tests

Further robustness tests are performed in this section. We used the Exponential General Autoregressive Conditional Heteroscedasticity (EGARCH) of Nelson (1991) to perform robustness checks. The EGARCH model of Nelson (1991) permits the asymmetric response of volatility to positive and negative returns. The mean and volatility between the portfolios of the stock market price are computed using the following equation:

$$R_{it} = \mu_i + \lambda_i R_t + \varepsilon_{i,t} \quad (3)$$

where, μ_i and λ_i are parameters to be estimated. R_{it} is the price of the stock market price at a time $t-1$. R_t is the price of the stock market price for all cryptocurrencies at time t . $\varepsilon_{i,t}$ an error term conditioned on the prior news Ω_{t-1} .

The conditional variance between the stock portfolios is captured by EGARCH. To complete the specification of the EGARCH estimation techniques, we state the conditional variance equations is written as:

$$\ln(\sigma_{i,t}^2) = \omega + \alpha_j \left[\frac{u_{t-1}}{h_{t-1}} \right] + \beta_j \ln(\sigma_{i,t-1}^2) + \chi_j \frac{u_{t-1}}{h_{t-1}} + \mu_{it} \quad (4)$$

In equation 4, ω indicates the permanent component of the conditional variance; α_j is the coefficient for the past historical news; β_j is the coefficient of past historical (lagged) conditional volatility; and χ_j is the coefficient that encapsulates the asymmetric – leverage effect. The news has asymmetric effects on volatility. In equation 4, the leverage exist if χ_j is < 0 , since χ_j is generally negative, negative shocks generate further volatility than it alternative positive shocks. An alternative lag selection may be selected using the Schwarz Information Criterion (SIC) rather than Akaike Information Criterion (AIC) used in our previous estimation. The SIC select two (2) lags rather than one (1) lag in estimation of the ANST-GARCH model described in equations (1) and (2).

[Table 6, about here]

The EGARCH is estimated with two lags and the empirical results are shown in Table 6. The results shown that there is no significant difference from the results shown in Table 5. Using both models, the study found a robust asymmetric mean reverting in cryptocurrency markets.

5.0 Conclusion and policy implications

This paper analyzed the asymmetric mean reversion and volatility in the cryptocurrency market from 20th April 2013 to 21st February 2023 using the ANST-GARCH model. Results from the study showed that asymmetric mean reverting exists in cryptocurrency markets and that strongly validates the asymmetric mean reverting hypothesis. The relevant explanation for the asymmetric mean reversion is the market overreaction theory. Findings show that positive shocks are more common than negative shocks in cryptocurrencies' return. An alternative model, EGARCH was used to perform robustness checks and the results are similar.

The results have important implications for investors, policymakers and for innovation purposes. Cryptocurrency continues to become an innovation within the financial system, gradually becoming conventional as a digital asset investment, technological infrastructure and a sharp development tool. Cryptocurrency has become so valuable in many economies in the world and as an alternative investment for investors. Fintech, bitcoin briefly hit highs up 15%, denominated in bitcoin, and one-year highs when denominated in dollars, suggesting rising inflows and speculation. This contributed to millions of short liquidations in recent times as prices pushed higher. Cryptocurrency is having a colossal impact on the conception of money as another kind of monetary currency for payment systems and money transfer payment in most economies in the world. Crypto asset could be used to diversify or reduce risk. Portfolio theory suggests that as an asset is added to a portfolio it reduces or diversifies the volatility.

Notwithstanding, for investors investing in crypto-assets can be challenging. Crypto-assets are, for the most part, unregulated products that presently linger beyond the scope of Security Exchange Commission (SEC). Crypt-assets are not traded on any documented or known stock exchange but somewhat are traded on predominantly unregulated crypto-exchanges

The study also showed that asymmetric volatility is crucially significant in cryptocurrency markets and that investors can implement risk management techniques and trading strategies to mitigate the effect. Notwithstanding, tests for asymmetries volatility showed the size effect of news, which is stronger for good news than for bad news. Future research could examine asymmetric reverting and volatility in cryptocurrency markets using an alternative model.

Conflict of Interest /ethical statements: There was no conflict of interest in the conduct of this research

Data Availability Statement: Data would be shared on request

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Table 1: DESCRIPTIVE STATISTICS

Variable	Mean	Median	Max.	Min	SD.	Skewness	Kurtosis	Jarque-Bera
BTC	0.755	0.034	87.122	-7.106	6.053	7.617 ^a	71.986 ^a	734726.4 ^a
ADA	0.718	0.953	7.337	7.11E-05	0.642	0.886 ^a	5.680 ^a	1519.9 ^a
BCH	0.760	0.112	24.462	-5.382	2.152	3.577 ^a	29.805 ^a	113306.8 ^a
DASH	0.368	0.002	9.291	7.81E-06	1.109	4.011 ^a	22.130 ^a	63345.5 ^a
DGB	0.233	0.047	20.750	-2.274	1.250	6.457 ^a	72.739 ^a	740499.2 ^a
DOGE	1.684	1.399	7.601	0.000	1.675	0.652 ^a	2.467 ^a	291849.0 ^a
ETH	0.350	0.063	11.973	-4.137	0.904	4.708 ^a	37.967 ^a	193037.5 ^a
LTC	0.359	0.011	8.985	8.41E-07	0.882	4.285 ^a	26.993 ^a	95556.0 ^a
MAID	0.052	0.002	1.584	1.61E-05	0.148	4.640 ^a	29.227 ^a	113936.4 ^a
SC	0.881	0.948	14.494	-7.461	2.469	-0.744 ^a	9.401 ^a	6358.2 ^a
USDT	-0.475	0.000	5.249	-4.752	1.665	-1.482 ^a	4.679 ^a	1707.7 ^a
VTC	0.054	0.007	1.660	0.000	0.124	4.861 ^a	37.883 ^a	193037.6 ^a
XEM	0.278	0.012	11.973	9.55E-05	0.795	5.923 ^a	50.919 ^a	358680.4 ^a
XLM	0.775	0.052	27.472	-4.256	2.221	3.834 ^a	30.807 ^a	122484.8 ^a
XMR	1.105	0.141	21.389	3.88E-05	2.429	3.488 ^a	18.024 ^a	40389.2 ^a
XRP	0.130	0.007	1.997	-0.206	0.319	3.423 ^a	15.932 ^a	31517.2 ^a
XVG	0.615	0.039	10.770	2.36E-05	1.127	5.284 ^a	36.513 ^a	181766.6 ^a

Our sample period span from 20th April 2013 to 21st February 2023 with 3533 observations for each variable. Obs. denotes observations. a, b. and c denote statistical significance at the 1%, 5% and 10% level, respectively.

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	BTC	ADA	BCH	DASH	DGB	DOGE	ETH	LTC	MAID	SC	USDT	VTC	XEM	XLM	XMR	XRP	XVG
BTC	1																
ADA	-0.13 ^a	1															
BCH	0.11 ^a	0.01	1														
DASH	-0.03	-0.03	0.10 ^a	1													
DGB	0.04	0.10 ^a	0.30 ^a	-0.12 ^a	1												
DOGE	-0.10 ^a	0.10 ^a	-0.16 ^a	0.35 ^a	-0.25 ^a	1											
ETH	0.60 ^a	-0.13 ^a	0.17 ^a	-0.14 ^a	0.07 ^b	-0.20 ^a	1										
LTC	-0.10 ^a	-0.21 ^a	-0.06 ^b	-0.13 ^a	0.08 ^b	-0.07 ^b	-0.09 ^a	1									
MAID	-0.07 ^b	0.02	-0.06 ^b	0.05	0.15 ^a	0.08 ^b	-0.11 ^a	-0.04	1								
SC	0.08 ^b	0.34 ^a	-0.42 ^a	0.05	-0.25 ^a	0.38 ^a	-0.08 ^b	-0.08 ^b	0.02	1							
USDT	0.00	-0.05	0.10 ^a	0.11 ^a	0.03	-0.30 ^a	0.11 ^a	-0.16 ^a	0.21 ^a	-0.02	1						
VTC	-0.11 ^a	-0.03	-0.15 ^a	-0.06 ^c	-0.06 ^c	0.01	-0.13 ^a	-0.03	0.24 ^a	0.01	0.02	1					
XEM	-0.06 ^c	0.11 ^a	0.03	-0.00	0.03	-0.05	0.33 ^a	-0.04	0.01	-0.02	0.10 ^a	-0.02	1				
XLM	0.00	-0.03	-0.18 ^a	0.25 ^a	-0.19 ^a	0.34 ^a	-0.07 ^b	-0.02	0.11 ^a	0.00	-0.27 ^a	-0.06 ^c	0.01	1			
XMR	-0.14 ^a	-0.21 ^a	-0.09 ^a	-0.14 ^a	-0.03	-0.20 ^a	-0.03	0.39 ^a	-0.12 ^a	-0.03	0.16 ^a	0.04	0.04	-0.13 ^a	1		
XRP	-0.05	-0.00	0.38 ^a	0.22 ^a	0.19 ^a	0.07 ^b	0.04	0.02	-0.04	0.06 ^c	0.07 ^b	-0.04	0.09 ^a	-0.34 ^a	-0.12 ^a	1	
XVG	-0.11 ^a	-0.35 ^a	-0.03	-0.10 ^a	-0.01	-0.22 ^a	-0.03	0.44 ^a	-0.10 ^a	-0.21 ^a	0.09 ^a	0.07 ^b	-0.02	-0.10 ^a	0.62 ^a	-0.12 ^a	1

Notes: Correlations coefficients between cryptocurrencies. a, b and c are significant at 1%, 10% and 5% respectively.

Table 3: Collinearity statistics

Variable	Collinearity statistics	
	Tolerance	VIF
BTC	0.587	1.254
ADA	0.556	1.272
BCH	0.456	2.006
DASH	0.724	1.205
DGB	0.735	1.201
DOGE	0.620	1.730
ETH	0.523	1.652
LTC	0.758	1.522
MAID	0.459	2.008
SC	0.568	1.820
USDT	0.756	1.542
VTC	0.658	1.893
XEM	0.735	1.554
XLM	0.498	2.003
XMR	0.563	1.423
XRP	0.739	1.516
XVG	0.687	1.810

Table 4: UNIT ROOT ANALYSIS

Variable	Augmented Dickey– Fuller		Phillips–Perron	
	Level	1 st Diff.	Level	1 st Diff.
BTC	-4.7638***	-14.4655***	-6.8572***	-73.4238***
ADA	-4.2283***	-22.3376***	-4.5119***	-72.7719***
BCH	-6.4347***	-10.7920***	-4.9704***	-64.3112***
DASH	-5.5994***	-8.3427***	-3.9146***	-67.0335***
DGB	-7.0888***	-11.1599***	-6.4747***	-62.5695***
DOGE	-3.1133**	-47.6501***	-3.2402**	-69.4864***
ETH	-6.5815***	-20.0703***	-7.3422***	-67.7285***
LTC	-4.7649***	-43.1439***	-5.9491***	-74.1229***
MAID	-6.2649***	-23.0512***	-7.2358***	-85.4014***
SC	-3.4298**	-20.4618***	-3.5135***	-59.9041***
USDT	-6.7593***	-27.1133***	-6.1093***	-72.6002***
VTC	-5.6623***	-35.2461***	-10.0751***	-95.7035***
XEM	-4.4228***	-19.6842***	-7.2591***	-54.9932***
XLM	-5.7812***	-24.3955***	-7.7895***	-88.7380***
XMR	-4.6569***	-10.5167***	-4.7141***	-66.2461***
XRP	-4.3189***	-15.1542***	-4.8245***	-82.8384***
XVG	-3.6271***	-12.6503***	-4.8694***	-63.3328***

Note that *, **, and *** denote the level of significance at the 10%, 5%, and 1% levels, respectively

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Table 5: Maximum likelihood estimates of ANST-GARCH model

$$R_t = \mu + [\phi_1 + \rho_1 D(\xi_{t-1})]R_{t-1} + \xi_t$$

$$h_t = [ao + a_1 \xi_{t-1}^2 + a_2 h_{t-1}] + [bo + b_1 \xi_{t-1}^2 + b_2 h_{t-1}] \{1 + \exp[-\gamma(\xi_{t-1})]\}^{-1}$$

	BTC	ADA	BCH	DASH	DGB	DOGE	ETH	LTC	MAID
μ	-0.07**	0.05**	0.04.	-0.08***	0.07**	-0.09***	0.05**	0.05**	0.04
	(0.03)	(0.03)	(0.04)	(0.02)	(0.03)	(0.01)	(0.03)	(0.02)	(0.04)
ϕ_1	0.37***	0.25***	0.31***	0.34***	0.19***	0.21***	0.35***	0.28***	0.18***
	(0.04)	(0.05)	(0.04)	(0.03)	(0.02)	(0.03)	(0.04)	(0.02)	(0.03)
ρ_1	-0.26***	-0.24***	-0.23***	-0.18***	-0.16***	-0.16***	-0.14***	-0.13***	-0.12***
	(0.04)	(0.03)	(0.02)	(0.01)	(0.00)	(0.02)	(0.05)	(0.03)	(0.02)
a_0	0.11**	0.09***	0.05**	0.08***	0.04	0.06***	0.05	0.09***	-0.12***
	(0.06)	(0.04)	(0.02)	(0.04)	(0.02)	(0.02)	(0.05)	(0.03)	(0.05)
a_1	0.33***	0.24***	0.29***	0.22***	0.32***	0.27***	0.36***	0.28***	0.35***
	(0.05)	(0.03)	(0.03)	(0.02)	(0.04)	(0.03)	(0.05)	(0.03)	(0.05)
a_2	0.86***	0.77***	0.94***	0.78***	0.65***	0.54***	0.95***	0.89***	0.82***
	(0.05)	(0.02)	(0.03)	(0.01)	(0.00)	(0.03)	(0.04)	(0.01)	(0.03)
b_0	0.09***	-0.5**	0.07**	-0.03	-0.04	-0.09***	0.06***	-0.01	-0.02
	(0.01)	(0.00)	(0.02)	(0.02)	(0.05)	(0.03)	(0.04)	(0.01)	(0.00)
b_1	0.22***	-0.25***	-0.29***	-0.26***	-0.28***	-0.21***	0.24***	0.23***	-0.27***
	(0.02)	(0.03)	(0.05)	(0.04)	(0.04)	(0.02)	(0.03)	(0.02)	(0.05)
b_2	-0.15***	-0.09**	-0.12***	0.10***	0.08***	-0.16***	-0.11***	0.15***	0.08***
	(0.04)	(0.02)	(0.05)	(0.04)	(0.01)	(0.03)	(0.02)	(0.03)	(0.01)
γ	92.20**	98.16**	87.24*	114.11***	78.36***	82.34***	88.10***	68.10***	78.24***
	(35.10)	(35.24)	(42.11)	(15.22)	(20.03)	(30.31)	(15.75)	(22.32)	(24.41)

Table 5: Continued

	SC	USDT	VTC	XEM	XLM	XMR	XRP	XVG
μ	0.06**	-0.01	0.05**	.02	0.08***	-0.03	0.07**	0.05**
	(0.02)	(0.00)	(0.02)	(0.01)	(0.02)	(0.01)	(0.03)	(0.03)
ϕ_1	0.27***	0.32***	0.31***	0.29***	0.25***	0.19***	0.23**	0.27***
	(0.04)	(0.03)	(0.05)	(0.03)	(0.02)	(0.04)	(0.06)	(0.05)
ρ_1	-0.12***	-0.11***	-0.10***	-0.14***	-0.13***	-0.14***	-0.12***	-0.10**
	(0.04)	(0.01)	(0.05)	(0.04)	(0.06)	(0.01)	(0.03)	(0.02)
a_0	0.02	0.05**	0.03	0.03	0.07**	-0.04	0.03	0.08***
	(0.02)	(0.02)	(0.01)	(0.01)	(0.03)	(0.01)	(0.02)	(0.03)
a_1	0.21***	0.23***	0.28***	0.31***	0.23***	0.36***	0.29***	0.31***
	(0.02)	(0.05)	(0.04)	(0.02)	(0.03)	(0.02)	(0.05)	(0.04)
a_2	0.76***	0.85**	0.58***	0.81***	0.59***	0.72***	0.68***	0.83***
	(0.02)	(0.05)	(0.04)	(0.05)	(0.04)	(0.01)	(0.03)	(0.02)
b_0	-0.09***	-0.07***	-0.04	0.03	-0.03	0.02	-0.08***	-0.05**
	(0.03)	(0.01)	(0.03)	(0.03)	(0.02)	(0.01)	(0.04)	(0.02)
b_1	-0.30***	-0.21***	-0.25***	0.16***	0.28***	-0.24***	-0.22***	-0.18***
	(0.05)	(0.02)	(0.03)	(0.01)	(0.03)	(0.02)	(0.02)	(0.01)
b_2	-0.10***	-0.12***	0.09***	-0.08***	0.16***	-0.14***	-0.12***	-0.10***
	(0.01)	(0.02)	(0.03)	(0.01)	(0.04)	(0.03)	(0.02)	(0.01)
γ	115.20***	85.15***	112.24***	78.67*	78.92***	88.51***	116.32***	92.84**
	(35.20)	(15.44)	(20.52)	(44.35)	(30.52)	(32.22)	(38.12)	(25.40)

Table 5: Continued

Diagnostics for ANST-GARCH model

	BTC	ADA	BCH	DASH	DGB	DOGE	ETH	LTC	MAID	SC	USDT	VTC	XEM	XLM	XMR	XRP	XVG
LB(18)	15.14.	20.23	23.44	20.14	19.15	25.10*	18.35	23.20	16.30	23.25	28.30**	16.20	23..12	21.86	19.20	20.24	18.30

LB ² (18)	10.25	8.35	18.65	17.45	12.30	19.22	24.20	8.15	18.68	21.28	22.98	16.15	14.78	15.65	22.55	11.20	16.88
Sign bias	-0.36	.034	0.39	-0.50	-0.30	0.15	-0.12	-.036	0.14	0.42	-0.16	0.32	-0.56	0.21	-0.35	-0.41	0.13
Negative size	-0.45	0.36	-0.65	-0.98	0.75	1.85*	0.56	-0.85	1.70*	0.69	0.85	1.75*	-0.95	0.92	-0.78	0.74	-1.85*
Positive size	1.90*	0.58	0.84	-0.68	0.45	1.89*	-0.78	-0.62	-0.95	0.86	-2.25**	0.68	0.65	1.90*	0.45	1.45	1.85*
Joint test	0.65	0.58	1.36	0.63	0.54	1.95	0.89	0.58	0.82	0.84	2.20*	0.86	1.23	1.89	1.92	1.52	2.30*

Note that *, **, and *** denote the level of significance at the 10%, 5%, and 1% levels, respectively. The standard errors are in parentheses. LB(18) and LB²(18) are the Ljung-Box test statistics testing for autocorrelation in the standardized residuals and standardized squared residuals of ANST-GARCH model up to the 18th lags.

Table 6: Maximum likelihood estimates of the EGARCH model

$$R_{it} = \mu_i + \lambda_i R_t + \varepsilon_{i,t}$$

$$\ln(\sigma_{i,t}^2) = \omega + \alpha_j \left[\frac{u_{t-1}}{h_{t-1}} \right] + \beta_j \ln(\sigma_{i,t-1}^2) + \chi_j \frac{u_{t-1}}{h_{t-1}} + \mu_{it}$$

	μ_i	λ_i	ω	α_j	β_j	χ_j	GED	$\alpha_j + \beta_j = 1$
BTC	-0.19*** (0.04)	0.42*** (0.02)	0.19*** (0.03)	1.10*** (1.12)	-0.56*** (0.04)	98.05*** (33.12)	3.65*** (0.72)	0.54***
ADA	0.07*** (0.05)	0.37*** (0.04)	0.12*** (0.03)	0.88*** (1.27)	-0.72*** (0.03)	88.01*** (30.20)	1.92*** (0.18)	0.16***
BCH	0.08*** (0.02)	0.27*** (0.05)	0.17*** (0.02)	0.75*** (0.05)	-0.49*** (0.03)	97.23*** (18.20)	4.10*** (0.81)	0.26***
DASH	-0.08*** (0.02)	0.25*** (0.03)	0.11*** (0.04)	1.16*** (0.04)	-0.64*** (0.04)	103.35*** (38.41)	3.89*** (0.75)	0.52***
DGB	-0.05*** (0.03)	0.33*** (0.04)	0.17*** (0.02)	0.85*** (0.03)	-0.55*** (0.04)	87.54*** (25.14)	1.39*** (0.15)	0.30***
DOGE	-0.08*** (0.04)	0.35*** (0.03)	0.09*** (0.03)	-0.09*** (0.04)	0.29*** (0.02)	78.25*** (28.26)	5.27*** (1.45)	0.20***
ETH	0.07*** (0.02)	0.40*** (0.02)	0.13*** (0.02)	0.95*** (0.02)	-0.38*** (0.03)	84.39*** (15.20)	2.54*** (0.58)	0.57***
LTC	0.06*** (0.01)	0.32*** (0.02)	-0.12*** (0.02)	1.12*** (0.04)	-0.62*** (0.02)	79.68*** (29.47)	1.65*** (0.17)	0.50***
MAID	0.02 (0.01)	0.28*** (0.03)	0.12*** (0.04)	0.87*** (0.03)	-0.73*** (0.04)	68.72*** (20.34)	3.87*** (0.73)	0.14***
SC	0.08*** (0.03)	0.27*** (0.03)	0.11*** (0.03)	0.92*** (0.04)	-0.35*** (0.01)	119.20*** (35.08)	2.38*** (0.62)	0.57***
USDT	-0.04 (0.03)	0.35*** (0.02)	0.09*** (0.03)	0.78*** (0.04)	-0.55*** (0.02)	98.38*** (27.84)	4.16*** (0.83)	0.23***
VTC	0.05** (0.02)	0.30*** (0.04)	0.12*** (0.01)	1.09*** (0.03)	-0.44*** (0.02)	115.80*** (32.26)	3.61*** (0.71)	0.65***
XEM	0.03 (0.04)	0.28*** (0.02)	0.11*** (0.03)	0.88*** (0.01)	-0.36*** (0.04)	85.67*** (16.30)	1.98*** (0.19)	0.52***
XLM	0.08*** (0.03)	0.32** (0.06)	0.12*** (0.04)	0.98 (0.03)	-0.65 (0.01)	79.10*** (25.22)	1.16*** (0.13)	0.33***
XMR	-0.04 (0.03)	0.29*** (0.03)	0.14*** (0.03)	-0.63 (0.02)	0.98 (0.04)	98.47*** (23.35)	3.57*** (0.70)	0.35***
XRP	0.09***	0.29***	0.08***	0.85	-0.39	119.26***	2.36***	0.46***

	(0.02)	(0.05)	(0.03)	(0.02)	(0.03)	(34.12)	(0.61)	
XVG	0.07***	0.28***	0.11***	0.83	-0.43	97.58***	2.48***	0.40***
	(0.01)	(0.04)	(0.02)	(0.04)	(0.03)	(29.22)	(0.64)	

Table 6: Continued

Diagnostics for EGARCH model

	BTC	ADA	BCH	DASH	DGB	DOGE	ETH	LTC	MAID	SC	USDT	VTC	XEM	XLM
LB (16)	19.32	20.21	23.78	24.10	21.56	25.92*	23.20	19.24	24.45	23.68	29.51**	19.65	24.25	21.10
LB ² (16)	11.24	10.56	17.64	19.10	16.58	20.45	19.14	12.57	24.05	19.84	23.10	12.58	19.32	14.48
Sign bias	-0.42	0.38	0.62	-0.38	0.48	-0.24	-0.28	0.36	0.54	-0.35	-0.69	0.47	-0.54	-0.41
Negative size	-0.48	0.37	-0.72	-0.98	0.78	1.88*	-0.74	0.58	1.72*	0.59	0.88	1.79*	-0.84	-0.73
Positive size	1.95*	0.84	0.92	-0.78	0.52	1.92*	-0.76	-0.84	-0.87	0.54	-2.48**	0.79	0.68	1.95*
Joint test	0.72	0.65	1.40	0.83	0.62	1.98	0.87	0.65	0.92	0.78	2.30*	0.86	1.27	1.90

Note that *, **, and *** denote the level of significance at the 10%, 5%, and 1% levels, respectively. The standard errors are in parentheses. LB(16) and LB²(16) are the Ljung-Box test statistics testing for autocorrelation in the standardized residuals and standardized squared residuals of ANST-GARCH model up to the 16th lags.

Information on each variable used in this study.

The Appendix 1 stipulates the perceptions of each cryptocurrency in terms of name, market capitalization, total supply and the date of issued

Name	Symbol	Market capitalization (USD)	Circulating supply (unit)	Date of issue
Bitcoin	BTC	738,501,403,305	19,556,350	03 rd Jan 2009
Ethereum	ETH	243,785,277,128	120,240,684	08 th Aug 2015
Litecoin	LTC	5,150,040,026	73,926,689	13 th Oct 2011
Nem	XEM	318,176,469	8,999,999,999	03 rd Apr 2015
Dash	DASH	354,041,080	11,560,000	15 th Feb 2014
Dogecoin	GOGE	11,505,839,289	142,030,856,384	06 th Dec 2013
Monero	XMR	3,036,585,760	18,364,475	21 st Apr 2014
Vertcoin	VTC	2,220,000	67,697,200	08 th Jan 2014
Verge	XVG	58,246,000	16,512,951,237	26 th Oct 2014
Digibyte	DGB	125,680,000	17,000,000,000	07 th Feb 2014
Stellar	XLM	3,332,697,988	27,993,969.734	05 th Aug 2014
Tether	USDT	89,105,689,319	89,079,709,234	07 th July 2014
Maidsafecoin	MAID	93,955,621	452,558,412	22 nd Apr 2014
Ripple	XRP	32,708,026,672	53,816,975,568	05 th Jun 2012
Bitcoin cash	DCH	4,378,397,132	19,572,281	01 st Aug 2017
Cardano	ADA	13,554,001,513	35,301,093,738	15 th Sep 2017
Siacoin	SC	259,574,763	51,480,000,000	03 rd Mar 2015